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Module III : Demonstration of Computer-Aided Design.

1. What is Computer-aided Design (CAD)? Discuss five major advantages of using digital CAD software (like auto CAD) over drawing by hand on a physical drafting board.

Ans⇒ Computer-Aided Design (CAD) refers to the use of Computer software to create, modify, analyze, and optimize engineering designs and blueprints. It replaces manual drafting with automated tools, precision geometry, and digital file management.

- Five major Advantages :-

1. Higher Accuracy and Precision:- CAD allows dimensions to be specified down to decimal fractions with zero manual drafting error, ensuring exact alignment and geometry.

2. Easy Modification and Editing:- Unlike physical drawings where mistake requires erasing or starting over, CAD allows designers to stretch, move, copy, or delete objects instantly.

3. Reusability and storage :- standard parts, symbols, and entire assemblies can be saved

as blocks and reused infinitely.

4. 3D Visualization and simulation :- CAD program enable 3D modeling, rendering, and assembly testing to check for interface and mechanical proforamce.

5. Efficient Documentation and Collaboration :- Drawings can be easily shared electronically via networks, plotted to any standard scale effortlessly.

2. Explain the three main ways to enter coordinate points in AutoCAD.

(i) Absolute Coordinates

(ii) Relative Rectangular Coordinates

(iii) Relative Polar Coordinates.

Ans ⇒ (i) Absolute Coordinates :- Points are entered relative to the origin point (0,0) of the drawing coordinate system. Specific a point 50 units along the X-axis and 30 units along the Y-axis from the absolute origin.

(ii) Relative Rectangular Co-ordinates :- Points are specified with reference to the previous point entered rather than the origin.

(iii) Relative Polar Coordinates:- points are specified based on a distance and an angle relative to the previous point.

3. Write down the exact command format you would type is AutoCAD to draw a line that is 100 units long of an angle of 45 degree, starting from the point 10,10.

Ans:) To draw a line of 100 units at an angle of 45° starting from the point (10,10), we get the line command and relative polar coordinates in AutoCAD.

Command : LINE

Specify first point : 10,10

Specify next point : @ 100<45

press enter to finish.

Explanation:

- starting point = (10,10)
- length of line = 100 units
- Angle = 45°
- Coordinates method = Relative Polar coordinates.

Calculation of end points:

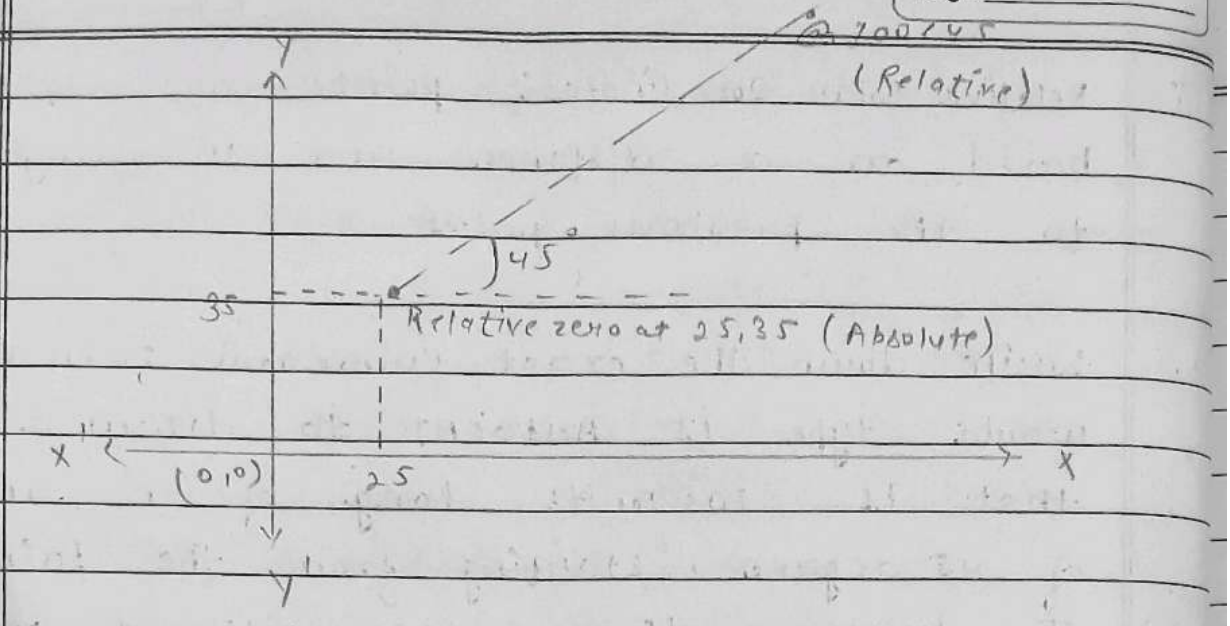
$x_2 = 10 + 100 \cos 45°$

$x_2 = 10 + 70.71 = 80.71$

$Y_2 = 10 + 100 \sin 45°$

$Y_2 = 10 + 70.71 = 80.71$

Therefore, the endpoint is approximately (80.71, 80.71).



4. What is the difference between an isometric projection and an isometric view? Also, define the term Isometric scale. Why do we need to use an isometric scale for an isometric projection, but not for an isometric view?

Ans =>

	Isometric projection	Isometric View
(i)	Uses an isometric scale.	Uses actual dimensions.
(ii)	Length are reduced.	Lengths are not reduced.
(iii)	Represent the object using true isometric projection principles.	Provides a convenient pictorial drawing with full dimensions.

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Isometric scale :- An isometric scale is a special scale used to obtain the reduced lengths required for an isometric projection.

- Why is an isometric scale required for an isometric projection?

In an isometric projection, the object is viewed in such a way that its lengths appear shortened. Therefore, an isometric scale is used to calculate the reduced lengths.

Q.5. Write down the step-by-step keyboard commands you would type in AutoCAD to draw a circle with a radius of 20 milimetre that is perfectly tanget to two lines meeting at a right angle.

Ans ⇒ To draw a circle of radius 20mm tanget to two perpendicular lines, we use the circle command with the TTR (tan, tan, radius) option.

Commands:

1. Type circle and press enter.
2. Type TTR and press enter.
3. Select the first line near the tanget point
4. Select the second line (horizontal line) near the tanget point.

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5. Enter the radius : 20
6. Press enter.

